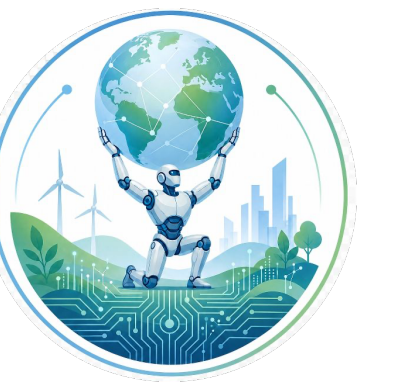




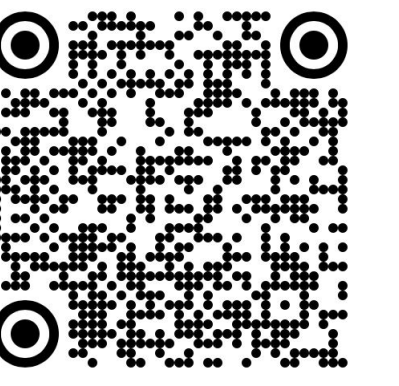
Winter Is Over for Continual Learning: From Continual Representation Learning to Continual Learning Agents

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1. Context and Perspective

- **Continual Learning (CL)** addresses a long-standing goal in AI: enabling machines to learn and adapt over time.
 - Much of the CL literature has focused on **updating model parameters while limiting forgetting**.
 - The rise of **LLMs** and **RAG**—offering strong zero-shot/few-shot generalization and access to external knowledge—made such continual updates appear less necessary, contributing to the “**CL winter**.”
 - **Agentic AI** brings the broader continual-learning problem back into focus: agents must operate over long time horizons in dynamic environments, integrate new knowledge, and adapt their behavior while preserving existing capabilities.
- These requirements can be supported **across the agent system**, motivating a shift from continual representation learning to **Continual Learning Agents**.

2. Continual Machine Learning: A Historical Arc

Early Era (1950–2012) Foundations of CL

- **Turing (1950)**: machine intelligence developing through education and experience
- **Newell & Simon (1972)**: symbolic systems with extensible knowledge
- **McCloskey & Cohen (1989)**: catastrophic interference in neural networks
- **1990s**: statistical models adapting to non-stationary data

Deep Learning Era (2013–2020) Continual Representation Learning

- Sequential training of deep neural networks
- **Replay, regularization, and architectural methods**
- Benchmarks and evaluation protocols

LLM Era (2021–2023) A Winter for CL

- Large-scale pre-training, in-context learning, and retrieval
- **Continual pre-training, continual fine-tuning**

LLM Agent Era (2023–present) Toward CL Agents

- Planning, memory, tool use, and multi-step interaction
- Long-horizon operation in dynamic environments
- **Continual knowledge accumulation** across interactions
- **Continual compositionality and orchestration** of models and tools

3. Continual Learning Agents

Continual Learning Agents are AI agents designed to pursue adaptation and consolidation as first-class objectives.

Adaptation

Responding efficiently to new tasks, user preferences, and changing environments

Consolidation

Accumulating and organizing knowledge and experience for efficient access over time

4. Adaptation and Consolidation Across the Agent System

Model

Updating model parameters

- **Continual pre-training** – updating general knowledge from new data
- **Continual fine-tuning** – adapting the model to new tasks or domains

External Memory and Knowledge

Maintaining and accessing evolving knowledge through external storage and retrieval

- Factual/declarative knowledge
- Interaction histories
- Reusable procedures distilled from past interactions

Compositionality and Orchestration

Selecting and coordinating evolving models and tools

- **Continual model routing** – selecting the right model as the model hub evolves
- **Continual model indexing** – refining model descriptions as the model ecosystem evolves

5. Open Challenges and Future Directions

System design

Determine which architectural principles enable sustained adaptation and consolidation.

Long-horizon evaluation

Measure knowledge accumulation, personalization, failure recovery, and adaptation over long horizons.

Safety

Prevent unintended behavioral drift and the accumulation of biased, erroneous, or misaligned knowledge over time.